B.Tech. (3rd year) V Semester (Computer Engg.) Examinations

Time: 1 hrs. **CEN - 501 Computer Architecture Sessional – I -2015** Max. Marks: 15

1	What do you mean by CSA? Draw the 64 bit CLA and calculate the worst case add time of 64 bit CLA?	5
2	Divide the 28 by 5 using **Non Restoring Division algorithm** and assume appropriate number of bits for numbers.	5
3	(a) Explain the Flynn's Classifications with example. (b) How Gustafson's- Barsis law differ from Amdahl's law.	3+2

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13BCS0013

B. Tech. (Computer Engineering) VII Semester
First Sessional Test 2015
COMPUTER NETWORK-I

Time: 1 Hour **MM: 15**

Q1: An ISP is granted a block of IP Addresses starting with 120.60.4.0/20. The ISP wants to distribute these blocks to 100 organizations with each organization receiving 8 addresses only. Design the subblocks and give the slash notations for each subblock. Find out how many addresses are still available after these allocations.

Q2: Neatly draw and explain all the fields of the Bluetooth data frame.

Q3: What are the necessary assumptions for Dynamic Channel Allocation? Explain briefly.

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ed: 14/09/15

1st Sessional Test

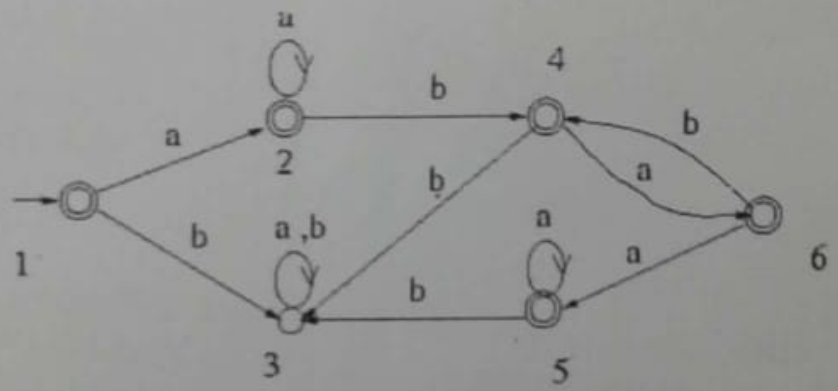
13BCS00013

Max Marks: 15

Automata Theory

B.Tech Comp 5th Sem, JMI

Q1) Minimize the following automaton 3+2 Marks



Also construct the R.E. for the newly obtained minimum automaton.

Q2) Construct a DFA which accepts all binary numbers which are multiples of 6. 3 Marks

Q3) Show that **(0+1)*01(0+1)*+1*0* = (0+1)*** 2 Marks

Q4) Verify the following statement: "***Subset of every regular language is not regular***". 2 Marks

Q5) Construct a DFA that accepts all strings including 'ε' that contain even number of a's & odd number of b's and do not contain the substring ab. 3 Marks

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Second Minor Examination 2014
B.Tech. (Computer Engineering) V semester
Microprocessor

Time:-One Hour

Attempt all questions. All questions carry equal marks

Max Marks:-15

Q1. How is an IO device interfaced with 8085 as I/O mapped I/O? Explain with help of suitable example.

Q2. Draw and explain timing diagram of LDA 2500H

Q3. How is INTR line of 8085 is used for handling (i) Single interrupt (ii)Multiple Interrupts ? Explain with help of suitable example and diagram.

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First Sessional Examination 2015

B.Tech. Computer Engineering V semester

13BCS50013

Microprocessor CEN-505

Marks:-*15*

Time:-One Hour

Attempt all questions. Marks are written in the margin.

Q1. Write a program in assembly language of 8085 to arrange a given set of integers stored at consecutive memory locations starting with address 2900H in descending order. First location contains number of item and following addresses contain items. Program should be well documented. 6 Marks

Q2. Write syntax and then explain the following instructions.

(i) XTHL (ii)LDAX (iii) SHLD 3 Marks

Q3.Write a program in assembly language of 8085 for 8X8 multiplication by shifting multiplicand. Program should be well documented. 3 Marks

Q4. Write a program in assembly language of 8085 for finding HCF of two nos. Program should be well documented. 3 Marks

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SET B

13BCS0013

Q1. WAP in C++ to design template classes such that they support following statements:

a. Ruppee <float> r1, r2;
b. Dollar <float> d1,d2;
c. d1 = r1;// Converts rupee to Dollar
d. r2 = d2;// Converts Dollar to rupee

Q2. WAP in C++ to overload the New & Delete Operator to perform allocation & de allocation of memory. Read numbers in that memory and find the largest of these.

DEPARTMENT OF COMPUTER ENGINEERING, JAMIA MILLIA ISLAMIA.
VI SEMESTER, I SESSIONAL FEB, 2016, OBJECT ORIENTED PROGRAMMING

Time: 1 hour Max. marks: 15 (3 + 8 + 2 + 2)

Answer all questions

1. Define the class Person which has name (char*) and age (int). Define the following constructors (a.) default (b.) with name as argument (assume age is 18) (c.) with name and age as argument. Also write a copy constructor to copy an object to another.
2. **Answer very briefly**
   i) Why does an overloaded "<<" returns an ostream& ? Is there any exception to this rule.
   ii) Why constructors do not return any value?
   iii) Why cannot a copy constructor take an object as a value parameter?
   iv) Why does a friend function always takes object as parameters?
   v) What is the difference between a copy constructor and an overloaded = operator.
   vi) What is the difference between overriding and overloading?
   vii) What is the difference between malloc() and new >
3. **Will the given program compile successfully. Give your answer with reasons.**

```
#include <iostream>
using namespace std;
 class Test
{
private:
  ~Test() {};
int main() { Test *t; }
```

4. Write a small program to show the use of a set_new_handler();

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Dated: 18/09/17

1st Sessional Test
Automata Theory
B.Tech Comp. 5th Sem, JMI

M. Marks: 15

Q1) Construct a DFA for the following expression: **a*((ab)* + c)*(ad)*** **(03)**

Q2) Consider two DFAs 'M' and 'N' as shown in given Fig. 1. The number of states in a minimal DFA that accepts the language L(M) ∩ L(N) is ___. Explain your answer with suitable reasons **(03)**

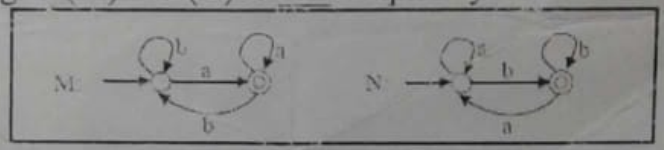


Fig.1

Q3) What is the minimum no of states in a DFA accepting the language L={w | w ε {0,1} *, where number of 0s and 1s in w are divisible by 3 and 5, respectively. Explain your answer with suitable reason. **(03)**

Q4) State if the following languages are regular or not using suitable reasons: **(03)**

i) L1 = {w ∈ {0,1}* | each string w has at least as many occurrences of (110)'s as (011)'s}.

ii) Let L2 = {w ∈ {0,1}* | each string w has at least as many occurrences of (000)'s as (111)'s}.

Q5) Minimize the following DFA using the Matrix Method. **(03)**

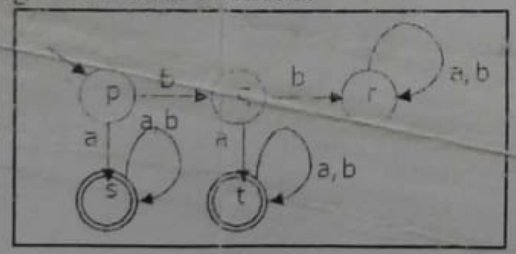


Fig.2

Validate your answers using suitable examples.

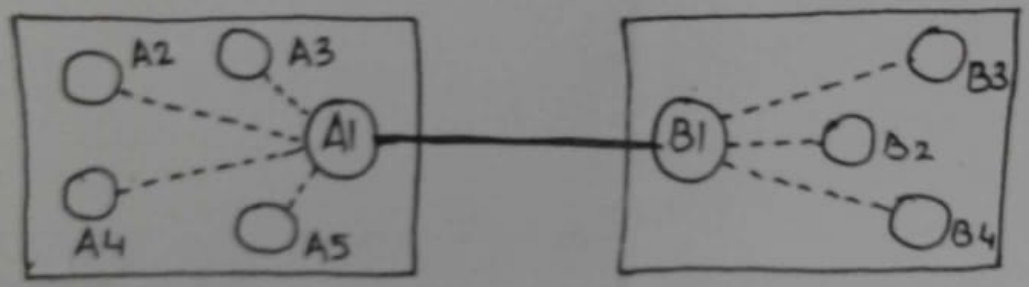
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B. Tech. (Computer Engineering) VI Semester, Second Sessional Test 2016

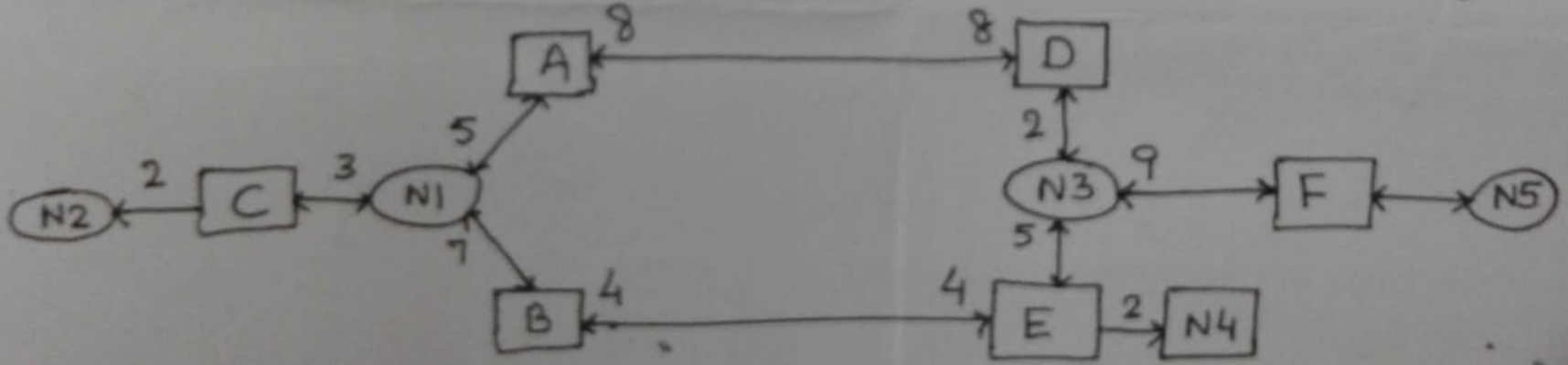
COMPUTER NETWORK II

Time: One Hour **MM: 15**

1) Show the BGP open message for router A1 in the figure



2) Show the link state update/network link advertisement for network N5 in figure



3) Discuss the IGMP messages.

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B.Tech. (Computer Engineering) VI Semester 2016

First Sessional Test 2016

COMPUTER NETWORK II

13BCS0013.

Time: 1 Hour MM:15(6+4+5)

1. Discuss the options in IP packets.
2. The following is the dump of a TCP Header in hexadecimal format.

05320017 00000001 00000000 500207FF 00000000

a) What is the source port address?
b) What is the destination port address?
c) What is the sequence no?
d) What is the acknowledgement no?
e) What is the length of header?
f) What is the type of segment?
g) What is the Window size?
h) What is the type of service?

All answers in decimal only.

3. A host with IP Address 130.23.43.20 and physical address B2:34:55:10:22:10 has a packet to send to another host with IP address 130.23.43.25 and physical address A4:6E:F4:59:83:AB (which is unknown to first host). The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.

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B.Tech. (Computer Engineering) VI Semester 2016

First Sessional Test 2016

COMPUTER NETWORK II

13BCS0013.

Time: 1 Hour MM:15(6+4+5)

1. Discuss the options in IP packets.
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B.Tech. (3rd year) V Semester (Computer Engg.) Examinations

Time: 1 hrs. CEN - 501 Computer Architecture Sessional – I Max. Marks: 15

Note: Attempt All Questions

1	Explain the NUMA model in details. Is NUMA is tightly coupled system?	3
2	Consider the following measurement made on two different machines running a given set of benchmark programs.	2

Instruction Category	No. of Instruction (in millions)	No of cycles per instruction
ALU	500	1
Load & Store	100	2
Branch	50	3
Others	50	4

What are the CPI, CPU time and MIPS rating for the machine A and B (assuming clock rate of 200 MHz)?

3	What is Moore's law? What are its consequences?	2
4	The memory access time is 2 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 3 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. Find the average memory access time (in nanoseconds) in executing the sequence of instructions?	2
5	Explain the shore's computer architectural classification scheme with example?	3
6.	State Amdahl's law? Suppose that we are considering an enhancement to the processor of a server system used for web serving. The new system is enhanced by using same type of processor to increase the computation time 10 times faster than the original processor. Assuming that the original CPU is busy with computational 40%of the time and is waiting for I/O 60% of the time, what is the overall speedup gained by incorporating the enhancement?	3

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B.Tech. (3rd year) V Semester (Computer Engg.) Examinations 2017

Time: 1 hrs. CEN - 501 Computer Architecture Sessional – II Max. M

Note: Attempt any 3 Questions

1	(a) What is Cache Coherence Problem? How can we resolve cache coherence problem in multiprocessor system. (b) Describe IEEE 754 standard for representing floating point numbers. Also mention the range of floating point numbers that are represented by IEEE 754 standard?
2	Draw the flow chart for Non-restoring algorithm. Divide the 23 by 4 using it and assume appropriate bits.
3	What is Booth Muitiplication? Show all steps for multiplying 12 and -12 using booth multiplication algorithm?
4	Design a hardwired controller for calculating the sum of the Fibonacci series up to n terms?

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SECOND SESSIONAL 2017

B. Tech- Computer Engineering 5th Semester

COMPUTER NETWORKS

Time: 1 Hour

Max. Marks: 15

Attempt all Questions:

1: What is Hidden Station and Exposed Station Problem? Discuss these problems with a suitable example. [4]

2: Compare and Contrast Datagram Subnet and Virtual Circuit Subnet. [4]

3: What is **Piconet** in Bluetooth Architecture? List various Layers in Bluetooth protocol architecture. [2]

4. Demonstrate with a suitable example How Distance Vector and Link State Routing Works? [5]

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B.Tech. (3rd year) V Semester, Computer Engg. Sessional – II **Year -2015**

Time: 1 hrs. **CEN - 501 Computer Architecture** Max.Marks: 15

Notes: Attempt All Questions

1 Design a 3-bit ALU that will meet the following specification without using multiplexer:

S_2	S_1	S_1	Output
0	0	0	A
0	0	1	A+1
0	1	0	A+B
0	1	1	A+B+1
1	0	0	$A+\overline{B}$
1	0	1	A-B
1	1	0	A-1
1	1	1	A

2 Design a micro programmed controller for finding sum of following series with reduced microinstruction length.

1-2+3-4+5-6+……n

(Note: USE single ALU for add and subtract)

3 Describe IEEE 754 standard for representing single and double precision floating point number format. Also calculate range of single and double precision floating point numbers. What do you mean by guard, round and sticky bit.

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Second Minor Examination 2015
Engineering) V semester

B. Tech. (Computer Engineering) V semester 2015
Second Sessional Test
COMPUTER NETWORK-I

MM:15 Time: 1 hour

Q1: Explain the process of Pulse Code Modulation.

Q2: Differentiate between Circuit, Message and Packet Switching.

Q3: Explain CRC with a suitable example.

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Second Minor Examination 2015
B.Tech. (Computer Engineering) V semester
Microprocessor

Time:-One Hour Max Marks:-15

Attempt all questions. All questions carry equal marks

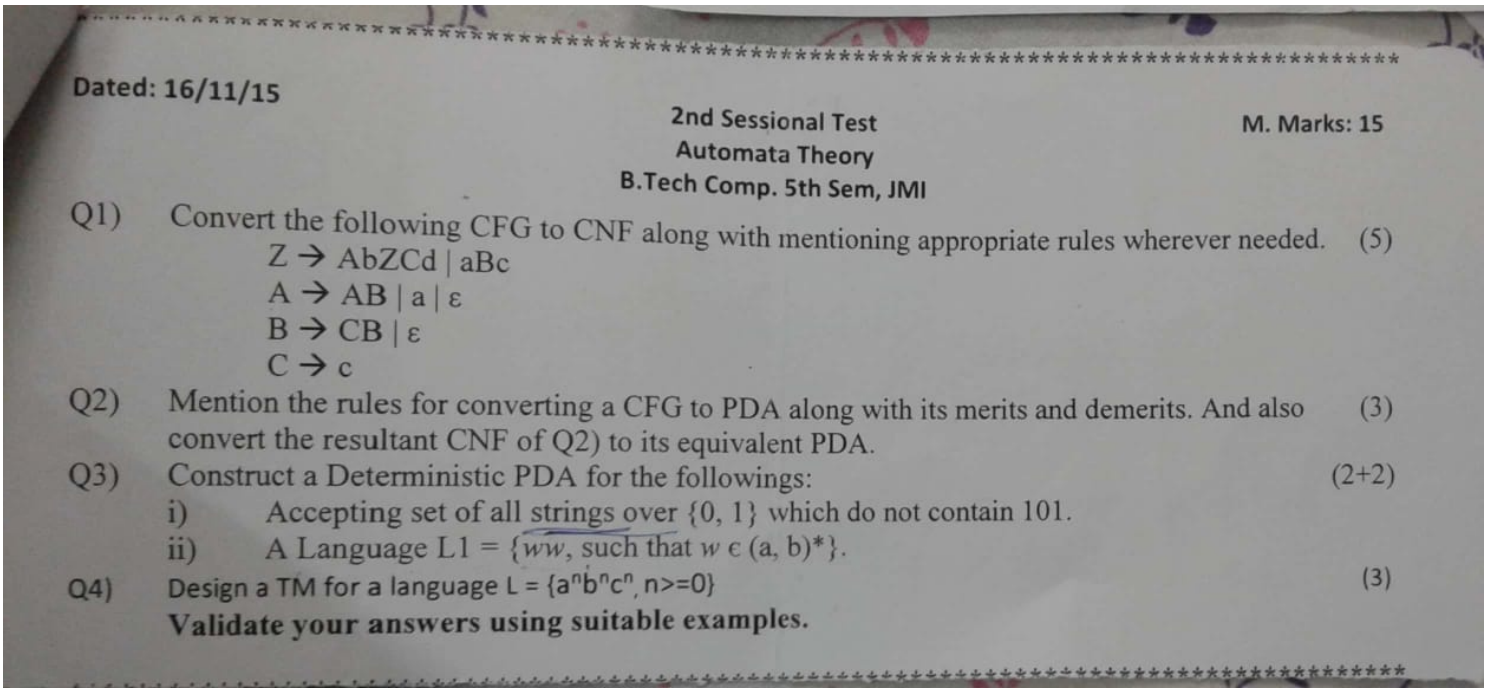
Q1. How an IO device interfaced with 8085 as memory mapped ~~is mapped~~ I/O? Explain with help of suitable example.

Q2. Draw and explain timing diagram of STA 2500H

Q3. How is INTR line of 8085 is used for handling (i) Single interrupt (ii)Multiple Interrupts ? Explain with help of suitable example and diagram.

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Dated: 16/11/15

2nd Sessional Test
Automata Theory
B.Tech Comp. 5th Sem, JMI

M. Marks: 15

Q1) Convert the following CFG to CNF along with mentioning appropriate rules wherever needed. (5)

$Z \rightarrow AbZCd \mid aBc$
$A \rightarrow AB \mid a \mid \varepsilon$
$B \rightarrow CB \mid \varepsilon$
$C \rightarrow c$

Q2) Mention the rules for converting a CFG to PDA along with its merits and demerits. And also convert the resultant CNF of Q2) to its equivalent PDA. (3)

Q3) Construct a Deterministic PDA for the followings: (2+2)

i) Accepting set of all strings over {0, 1} which do not contain 101.

ii) A Language L1 = {*ww*, such that *w* ϵ (a, b)*}.

Q4) Design a TM for a language L = $\{a^n b^n c^n, n>=0\}$ (3)

Validate your answers using suitable examples.

B.Tech (Computer Engineering) Semester- V Sessional II 2017
Object Oriented Programming CEN-504

Time: 1 Hou

Maximum Marks: 15

Q1. ~~(a)~~ What are the operators in C++ that cannot be overloaded? (1 mark)

~~(b)~~ What do you mean by the term 'Generic Programming'? Differentiate between function template and class template with suitable example? (3 marks)

(c) What are virtual functions? Give an example to highlight their use in implementing run time polymorphism in C++? (3 marks)

Q2. ~~(a)~~ Justify the statement "Java is platform independent". (1 mark)

~~(b)~~ What is the use of final keyword in java? Show with example. (2 marks)

~~(c)~~ What are the two ways of using super keyword? class/fun^n. (2.5 marks)

~~(d)~~ Differentiate between abstract class and interface. (2.5 marks)

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B.Tech (5th Semester) Computer Engineering

Object Oriented Programming – 1 (2017)

Time : 1 Hour

Max. Marks : 15 (5+5+5)

Q.1. a) Why there is no return type in a constructor?

b) What is the implicit declaration of "this" pointer? How can use "this" pointer efficiently.

c) What is an array of pointers to objects and how it is used?

d) You are given the following class containing the following declarations.

```
class MATRIX {
                int MAXROW, MAXCOL; // number of rows and cols
                int **p; // pointer to a 2-D array
        public : Write a constructor to initialize a two dimensional array dynamically.
                };
```

Q2. (a) What is an inline function? Whether it has some advantage or disadvantage? Discuss.

(b) What is a copy constructor? Is it necessarily invoked while copying objects? Under what all conditions does a copy constructor invoke. Discuss with the help of a complete program.

Q3. (a) How are constructors and destructors executed in multilevel inheritance? Explain you answer with suitable example.

(b) What do you mean by a virtual base class in C++? What is its significance? Explain with a complete program.

virtual public base 2
public virtual base 1

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FIRST SESSIONAL 2017

B. Tech- Computer Engineering 5th Semester

Time: 1 Hour | COMPUTER NETWORKS | Max. Marks: 15

Attempt all Questions:

1: Which of the OSI Layers handles each of the following? [1+1=2]

(a) Dividing the transmitted bit stream into frames.

(b) Determining which route through the subnet to use.

2: Discuss the role of ATM adaption layer (AAL) in ATM reference Model briefly. [2]

3. What is the essential difference between message switching and packet switching? [3]

4. Differentiate between Acknowledged connectionless service and Acknowledged connection - oriented service. [2]

5. Explain with a suitable example A simplex stop- and- wait protocol. [3]

6. What is pure ALOHA and Slotted ALOHA? Consider the delay of pure ALOHA versus slotted ALOHA at low load. Which one is less? Explain your answer. [3]

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Total Pages : 2

Your Roll No. ....13BCS-0013.

CEN-503

B. Tech. Computer Engineering.

V^th Semester Examination, 2015

Computer Network-I

Paper No. : CEN-503

Time : 3 Hours

Maximum Marks : 60

(Write your Roll No. on the top immediately on receipt of this question paper.)

Note : Attempt any two parts from question 1 to 4. Question 5 is compulsory.

1. (a) A Low Pass signal with a bandwidth of 300 Khz is sampled using 1024 levels of quantization. **06**

   (i) Find the bit rate of the digitised signal.

   (ii) Find the SNR in db for this signal.

   (iii) Find the PCM bandwidth of this signal.

   (iv) Given a channel with an intended capacity of 20 Mbps, the bandwidth of the channel is 3 Mhz. Find SNR to achieve this capacity. Assuming the same channel capacity find the number of signal levels using nyquist formula.

   (b) Differentiate between Packet Switching, Circuit Switching and Message Switching. **06**

   (c) Discuss PCM. Twenty Four voice signals are to be multiplexed and transmitted over twister pair. What is the bandwidth required for FDM? Assuming a bandwidth efficiency of 1 bps/Hz, What is the bandwidth required for TDM using PCM?

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2. (a) An ISP is granted a block of addresses starting with 150.80.0.0/16. The ISP wants to distribute these blocks to 1000 customers as follows : 06

(i) The first group has 20 medium type businesses; each needs 128 addresses.

(ii) The second group has 400 small businesses; each needs 16 addresses.

(iii) The third group has 2000 households; each needs 4 addresses.

Design the subblocks and give the slash notations for each subblock. Find out how many addresses are still available after these allocations.

(b) Consider a space division switch with 1000 inputs and outputs. What is the total number of cross points in each of the following cases ? 06

(i) Using one single crossbar.

(ii) Using a multistage switch based on cross criterion.

(c) What is single mode and multimode optical fibre ? Discuss advantages and disadvantages of Optical fibre. 06

3. (a) Draw the HDLC Frame Format and discuss its fields. 06

(b) Discuss how Selective Repeat is better than Stop and Wait ARQ. 06

(c) Differentiate between Token Ring and CSMA/CD. 06

4. (a) Neatly draw the Bluetooth Data Frame and explain its fields. 06

(b) Discuss the MAC protocol for Wireless LAN. 06

(c) Discuss the Channel Allocation problem. 06

5. (a) Differentiate between Link State Routing and Distance Vector Routing. 06

(b) Explain the terms Unicast, Multicast and Broadcast with suitable example for each case. 06

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Q1. How an i
suitable exam

Time: 1

Total Pages : 3

Your Roll No. 13.BCS-0013.

CEN-502

B.Tech Computer Engineering,

Vth Semester Examination, 2015

Automata Theory

Paper No. : CEN-502

Time : 3 Hours

Maximum Marks : 60

(Write your Roll No. on the top immediately on receipt of this question paper.)

Note : Attempt all questions.

1. (a) Explain Deterministic Finite Automaton with suitable example. **(6 marks)**

Also design a Finite Automaton to accept all strings over {a, b} containing oven number of a's and odd no. of b's.

(b) Construct a DFA for the following languages and validate your answers : **(6 marks)**

(i) ab(a + ab + abc)*cd

(ii) 11(1 + 11)*0*1*

OR

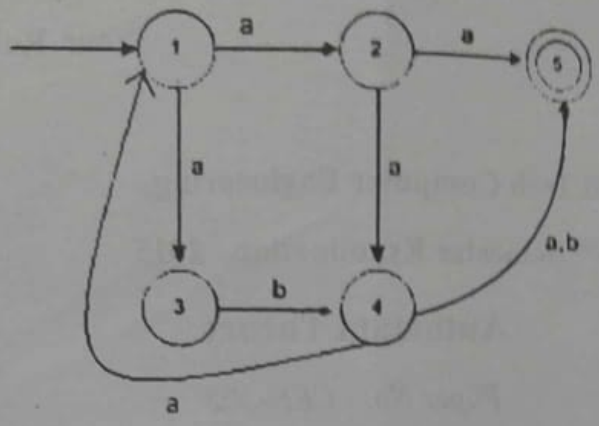
1'. (a) What are transition diagrams? How is an automaton related to a transition diagram? Explain using suitable examples. **(6 marks)**

(b) Briefly describe Mealy and Moore machines. Also construct a Mealy Machine that output '1' whenever it receives a pair 10 in a string with $\Sigma = (0, 1)$. **(6 marks)**

2. (a) State and prove the Arden's theorem. Also explain the usefulness of this theorem. **(6 marks)**

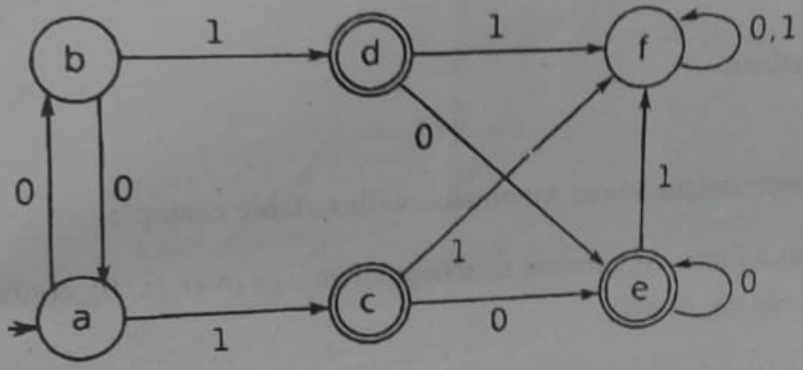
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(b) Convert the following NFA to DFA. **(6 marks)**



OR

2'. (a) Minimize the following DFA : **(6 marks)**



(b') Construct a regular expression for the minimum DFA obtained as an answer of Q2'a. **(6 marks)**

3. (a) Using pumping Lemma, show whether the following languages are regular or not : **(6 marks)**

(i) $L1 = \{ww : w \in \{a,b\}^+\}$

(ii) $L2 = \{1^x 1^z : x \text{ is even}\}$

(b) Construct a CFG for a language accepting all strings over {0, 1} which must contain a substring 110. Also define ambiguity of CFG using suitable examples. **(6 marks)**

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4. (a) Convert the following CFG to CNF (6 marks)

$X \rightarrow XYZ \mid XxYyZz \qquad Y \rightarrow Yy \mid XZ \qquad Z \rightarrow \varepsilon$

$X \rightarrow xy \qquad Y \rightarrow \varepsilon$

Mention all the steps and the corresponding rules used at appropriate positions.

(b) Mathematically define a PDA. Also construct a PDA for a Language that accepts set of all strings over {a, b} which do contain equal no. 'a' and 'b'. Validiate your answers. (6 marks)

5. (a) Construct a Turing Machine to accept a Language $L = \{a^n b^n c^n : n>=0\}$. (6 marks)

Also design a TM for the operation: $\sqcup\sqcup, w\sqcup\sqcup, \rightarrow \sqcup\sqcup w\sqcup\sqcup w^R\sqcup\sqcup$, where w^R stands for reverse of the string w. (6 marks)

(b) Define Primitive Recursive functions. Show that the f(x, y) = 2x + y is primitive recursive. (6 marks)

OR

5' (b) Explain following terms :

(i) Enumerable Languages

(ii) NP Complete

(iii) Church thesis

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standard for representing si
and mention the range of it. Ad
ormat. What do you mean by gu

t will meet the following specif

Signals		Output
S_1	S_1	
0	0	A
0	1	A+1
1	0	A+B
1	1	A+B+1
0	0	A–B–1
0	1	A–B
1	0	A–1
1	1	A
0	x	A AND B
1	x	A XOR B
0	x	A OR B
1	x	NOT A

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OR

(a) What do you mean by Booth multiplication algorithm and show all the steps for multiplication of (–18) and (5) using it. (Note : Take appropriate number of bit) 6

(b) Draw the flow chart for Non-Restoring Division algorithm. Divide 28 by 5 using it and assume appropriate number of bits for numbers. 6

(a) Design a simple computer to support the following instruction : 9

LDA Address

LDI OPRD

MOV R

ADD R

(b) Design an Array Multiplier that multiplies two 4 bit numbers. Use AND gates and full adders. 3

4. (a) Consider the following algorithm : 6

Declare register A[8], B[8], S[8], I[8], C[8]

Step 1 : A ← 0, S ← 00000000;

Step 2 : B ← 00000001;

Step 3 : I ← Inbus;

Step 4 : C←A+B;

Step 5 : S←S+C;

Step 6 : A←B;

Step 7 : B←C, I←I-I;

Step 8 : if I<>0 then go to step 4

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Step 9 : Outbus←s;

Step 10 : Go to halt

Design a micro programmed controller that implements this algorithm with reduced microinstruction length.

(b) Design a Hard wired controller for finding sum of following series (6)

5-10+15-20+25-30+....n

(Note: where n is multiple of 5 and USE single ALU for add and subtract)

5. (a) (i) What do you mean by shared memory and message passing systems? 3+3=6

(ii) What are data hazards ? A 5 stage pipeline adder is to be designed. The time taken by each of the pipeline stages are 12ns, 4ns, 8ns, and 6ns. What is the time to add 150 pairs of operands?

(b) Briefly explain any ***three*** of the following : (2*3=6)

(i) VLIW processor

(ii) Superscalar processor

(iii) Difference between Array Processor and Vector Processor

(iv) Nanomemory

OR

5'. (a)' Design 8-bit CLA and compare it with 8-bit CPA. Also calculate the worst case add time for 64 bit CLA and CPA. 6

(b)' Explain the Amdahl's law and Gustafson's Barsis Law. What are the shortcomings of both the algorithms? Suppose a program consists of 0.45 fractional parts of sequential code which is running on parallel computer having 12 processors and each of them have equal speed which have greater speed up and why ? Give reasons. 6

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Total Pages : 2

Your Roll No. ....13BCS-0013.....

CEN-503

B. Tech. Computer Engineering.

Vth Semester Examination, 2015

Computer Network-I

Paper No. : CEN-503

Time : 3 Hours

Maximum Marks : 60

(Write your Roll No. on the top immediately on receipt of this question paper.)

Note : Attempt any two parts from question 1 to 4. Question 5 is compulsory.

1. (a) A Low Pass signal with a bandwidth of 300 Khz is sampled using 1024 levels of quantization. **06**

   (i) Find the bit rate of the digitised signal.

   (ii) Find the SNR in db for this signal.

   (iii) Find the PCM bandwidth of this signal.

   (iv) Given a channel with an intended capacity of 20 Mbps, the bandwidth of the channel is 3 Mhz. Find SNR to achieve this capacity. Assuming the same channel capacity find the number of signal levels using nyquist formula.

   (b) Differentiate between Packet Switching, Circuit Switching and Message Switching. **06**

   (c) Discuss PCM. Twenty Four voice signals are to be multiplexed and transmitted over twister pair. What is the bandwidth required for FDM? Assuming a bandwidth efficiency of 1 bps/Hz, What is the bandwidth required for TDM using PCM?

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2. (a) An ISP is granted a block of addresses starting with 150.80.0.0/16. The ISP wants to distribute these blocks to 1000 customers as follows : 06

(i) The first group has 20 medium type businesses; each needs 128 addresses.

(ii) The second group has 400 small businesses; each needs 16 addresses.

(iii) The third group has 2000 households; each needs 4 addresses.

Design the subblocks and give the slash notations for each subblock. Find out how many addresses are still available after these allocations.

(b) Consider a space division switch with 1000 inputs and outputs. What is the total number of cross points in each of the following cases ? 06

(i) Using one single crossbar.

(ii) Using a multistage switch based on cross criterion.

(c) What is single mode and multimode optical fibre ? Discuss advantages and disadvantages of Optical fibre. 06

3. (a) Draw the HDLC Frame Format and discuss its fields. 06

(b) Discuss how Selective Repeat is better than Stop and Wait ARQ. 06

(c) Differentiate between Token Ring and CSMA/CD. 06

4. (a) Neatly draw the Bluetooth Data Frame and explain its fields. 06

(b) Discuss the MAC protocol for Wireless LAN. 06

(c) Discuss the Channel Allocation problem. 06

5. (a) Differentiate between Link State Routing and Distance Vector Routing. 06

(b) Explain the terms Unicast, Multicast and Broadcast with suitable example for each case. 06

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15BCS-74
Roll No.

CEN-501

B. Tech. Computer Engineering V Semester Examination, 2017
COMPUTER ARCHITECTURE
Paper No. CEN-501

MM: 60

Time: Three Hours
(Write your Roll No. on the top immediately on receipt of this question paper.)
Note: Attempt All Questions. Assume suitable missing data, if any.

Q1 a) Describe Flynn's and Handler's computer architectural classification scheme with example. 6

b) Consider the following measurement made on two different machines running a given set of benchmark programs. 6

Instruction Category	No. of Instruction (in millions)		No of cycles per instruction	
	Machine A	Machine B	Machine A	Machine B
ALU	8	10	1	1
Load & Store	4	8	3	2
Branch	2	2	4	4
Others	4	4	3	3

(a) Define CPI, CPU Time?
(b) What are the CPI, CPU time and MIPS rating for the machine A and B (assuming a clock rate of 200 MHz)? Which Machine is better and why?

OR

b)' (i) Describe IEEE 754 standard for representing floating numbers. Also mention the range of floating point numbers that are represented by IEEE 754 standard. Also represent the X=5.75 in IEEE 754 standard for representing single and double precision floating numbers? 5

(ii) What are the consequences of Moore's law? 1

Q2 a) What do you mean by Booth multiplication algorithm and show all the steps for multiplication of (-29) and (8) using it (assume appropriate no of bits)? 6

b) Draw the flow chart for Non-restoring algorithm. Divide 28 by 5 using it and assume appropriate no of bits? 6

OR

a) Design 8-bit CLA and compare it with 8-bit CPA. Also calculate the worst case add time for 64 bit CLA and CPA? 6

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b) What is Braun's Multiplier? Design a Braun's Multiplier for 5X5 multiplication. Also calculate the speedup of the designed multiplier. 6

Q3 a) What is addressing modes? Define Base Register addressing mode, indexed address mode and indirect addressing mode? What are the advantages/disadvantages of using large no addressing modes in CPU design? 6

OR

a)' Consider the following algorithm: 6

```
Declare register A[8], B[8], S[8], I[8], C[8]
Step 1: A ← 0, S ← 00000000;
Step 2: B ← 00000001;
Step 3: I ← Inbus;
Step 4: C ← A+B;
Step 5: S ← S+C;
Step 6: A ← B;
Step 7: B ← C, I ←I-1;
Step 8:if I<>0 then go to step 4
Step 9: Outbus ← S;
Step 10: Go to halt
```

Design a micro-programmed controller that implements this algorithm?

b) Design a hardwired controller for calculating the sum of the Fibonacci series up to n terms? 6

Q4 a) What do you mean by Shared memory? Briefly discuss different shared memory architecture models? 6

b) (i) What is Cache Coherence Problem? How can we resolve cache coherence problem in multiprocessor system? 4+2=6

(ii) The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 3 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 200 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. Find the average memory access time (in nanoseconds) in executing the sequence of instructions?

Q5 a) (i) What is branch prediction technique? Also explain how is it emplemented with the help of suitable diagram and example. 3+3=6

(ii) A 4-stage pipeline has the stage delays as 150, 120, 160 and 140 nanoseconds respectively. Registers that are used between the stages have a delay of 5 nanoseconds each. Assuming constant clocking rate, what will be the total time taken to process 1000 data items on this pipeline?

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	I1	I2	I3	I4
S1	2	1	1	2
S2	2	1	3	1

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CEN-505Roll No.....

B.Tech.(Computer Engineering)

Vth Semester Examination, 2017

COMPUTER NETWORKS

Time : 3 Hours | CEN-505 | Max. Marks: 60

(Write your Roll No. on the top immediately on the receipt of this Question paper.)

Note: (i)Attempt ALLQuestions.
(ii) Question No. 1 is Compulsory.

1. Attempt any **FOUR** of the following questions: [4x3=12]
   (a) List down goals and applications of Computer Networks
   (b) What are the liabilities of the Network Layer and Application Layer?
   (c) Differentiate between Circuit Switching & Packet Switching.
   (d) Write about Peer to Peer Processing.
   (e) What is geosynchronous satellite? writeabout VSATs- (Very Small Aperture Terminals).
   (f) Explain Repeaters, Bridges, Switches.

Attempt any 2 part from Question no.2 to Question, no.5.

2.(a)Write about ALOHA Protocols. Differentiate between Pure ALOHA and Slotted ALOHA with a neat Diagram. [6]

(b) Explain using an example, How bit stuffing is used to preserve frame boundaries when transmitting binary data at the Data Link layer of the protocol Stack. List all layers in TCP/IP Protocol Scuite. [6]

(c) List various services provided by Data Link layer to the Network Layer and differentiate between Unacknowledged connectionless service and Acknowledged connection-oriented service. [6]

3.(a)Explain with a suitable example how ARQ.(Automatic Repeat reQuest) orPositive Acknowledgement with Retransmission protocolworks? [6]

(b) Write about the following protocols[6]

- IEEE 802.11
- CSMA/CA

P.T.

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4. (a) Write about the following Mutliplexing Techniques: [6]

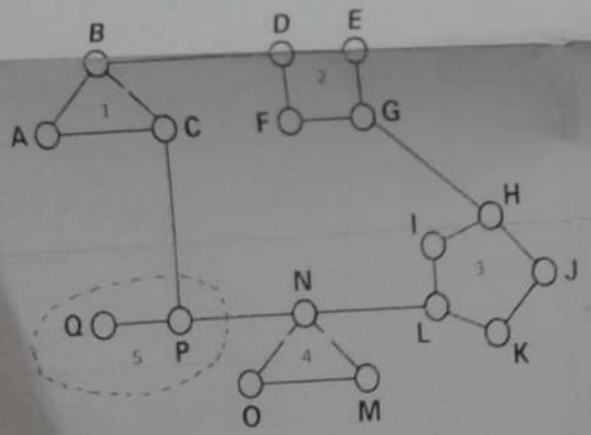
- TDM
- CDMA

4. (a) Discuss in detail Network Layer Design issues. [6]

(b) Demonstrate with an example How Distance Vector Routing and link state routing works? [6]

(c) Compare and contrast Datagram subnet and Virtual Circuit subnet. [6]

5. (i) Draw a Hierarchical table for A using Hierarchical routing if the full table for A and 2- level hierarchy with five regions is given asfollows: [3]



(ii) List few General Principles of Congestion Control. [3]

(b) List various Techniques for achieving Good Quality of Service(QoS)requirements and briefly discuss Buffering techniques. [6]

(C) Write short on the following: [6]

- Internetworking.
- Congestion Control in Virtual Circuit Subnet

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Roll No. 15BCS0074

-302

B.Tech. (Computer Engineering) V Semester Examination, 2017

Object Oriented Programming
Paper No. CEN-504
Time: Three Hours
Maximum Marks: 60

(Write your Roll No. on the top immediately on receipt of this question paper)

Note: Attempt all questions.

1. (a) (i) What are the basic concepts of object-oriented programming? Define polymorphism and encapsulation. 1

(ii) What is a reference data type? How is it different from a pointer data type? 1

(iii) You are given a class containing the following declarations. 2.5

```
class MATRIX {
        int MAXROW, MAXCOL; // number of rows and cols
        int **p; // pointer to a 2-D array
public: Write a constructor to initialize a two dimensional array dynamically.
        };
```

(b) Differentiate: default constructor, copy constructor, dynamic constructor, conversion constructor and explicit constructor? 2.5

(c) With the help of a small complete program show the method of passing parameters from the derived class constructor to a base class constructor? 3

(d) In C++, protected keyword is specifically useful in which situation ? Show with a small piece of code. 2

2. (a) Design two classes, Centigrade and Fahrenheit, such that they support the following statements : 5

```
Centigrade c1, c2;
Fahrenheit f1, f2;
c1 = f1;        // Fahrenheit to Centigrade
f2 = c2;        // Centigrade to Fahrenheit
```

Write a program in C++ that does such conversions according to the following formula :

C/5 = (F - 32)/9

(b) Write a program in C++ to perform multiplication of an integer with an object. Use friend function. 4

(c) What are static data members and static member functions and how are they different from normal data and functions? 3

$\frac{C}{5} \times 9 + 32 =$

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Q3. (a) What do you mean by operator overloading? Write a program to show the overloading of ***new*** and ***delete*** operators. 5

(b) What is a virtual base class and why it is used? Write a program to illustrate the concept. 3.5

(c) Explain how run time polymorphism is realized using virtual function in C++. 3.5

OR

Q3' (a) Why is it necessary to overload an operator? Define a class String. Use overloaded == operator to compare two strings. 5

(b) What is a pure virtual function and what are its uses? 2

(c) Write a program in C++ to find maximum of two data items irrespective of type of items(s) using templates. 3

(d) How many arguments are required in the definition of an overloaded unary operator? 2

Q4. (a) In the following Java statement, explain the significance of each keyword : 2

public void static main (String args[]);

(b) Critically comment on the validity of following statements pertaining to Java programs : 2

(i) 'Static method cannot be overridden.'

(ii) 'Abstract class is faster than interface class.'

(c) With the help of suitable example, explain the differences between a class, an abstract class and an interface in Java. Write a program to show the use of ***interface*** in implementing run-time polymorphism. 5

(d) What is the difference between ***final***, ***finally*** and ***finalize***? 3

OR

Q4' (a) Explain why multiple inheritance is not possible in Java? What is the different approach to achieve it? Show with a complete program. 6

(b) What is user-defined exception? Write a program in Java by using ***throw*** and ***throws*** clause in an appropriate method to find maximum roots of the equation $ax^2 + bx + c = 0$. 6

Q5. (a) What is the difference between Error and Exception? Name any five common exceptions? Write the code to handle multiple type of exceptions occurring in a single ***try*** block. 6

(b) What do you mean by Thread synchronization? How do we implement it? Show with an example code. 6

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Roll No. 12CS6...

Code No:CEN-501

Name of Examination: B.Tech (Computer Engg.) V Semester Examination, 2014

Subject/ Name of course: Computer Architecture

Paper No. CEN-501

Time: Three Hours

Maximum Marks: 60

(Write your Roll No. on the top immediately on receipt of this question paper)

Note: Attempt all questions. Assume suitable data wherever required and state them clearly.

Q1. (a) (i) Explain Shore's computer architectural classification scheme in details with example. 4

(ii) What are the consequences of Moore's law? 2

(b) (i) Represent 28.75 in IEEE 754 standard for representing floating numbers in single and double precision format. What is range of single and double precision format? 4

(ii) Consider the following measurement made on a machine running a given set of benchmark programs. 2

Instruction Category	No. of Instruction	No of cycles per instruction
ALU	500	1
Load & Store	100	2
Branch	50	3
Others	50	4

CPI = IC

CPU = CPI × I

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(b)' Draw the flow chart for Non Restoring Division algorithm. Divide the 29 by 5 using it and assume appropriate number of bits for numbers.

Q3 Design a simple computer to support the following instruction:

LDA Address
MVI A, Data
MOV A, R

Q4. (a) Design a hardwired controller for the calculating the sùm of following series up to N terms:

1 - 2 + 3 - 4 + 5 - 6 +. . . . . N

(b) Consider the following algorithm :

Declare register S[8], I[8], N[8], D[8]
Step 1 : S←0, I←0; N← Inbus;
D←00000110;
Step 2: If N[0]<> 0 then goto step 4
Step 3: I← N goto step 6
Step 4: N←N-1;
Step 5: I←N;
Step 6: S←S+I;
Step 7: I←I-D;
Step 8: If I<>0 then go to step 6
Step 9: Outbus←S;
Step 10: Go to halt

Design a controller that implements this algorithm using microprogrammed approach with reduced microinstruction length.

OR

Q5. (a) Explain the different types of data Hazards with an example?

A 5 stage pipeline adder is to be designed. The time taken by each of the pipeline stages are 10ns, 5ns, 8ns and 6ns.

i. What should be the clock frequency to drive the adder?
ii. What is the time taken to add 150 pairs of operands?
iii. If 45 operands are zero (at random), what is the time taken to add 150 pairs of operands?
iv. What is the efficiency of pipeline addition if all the operands are non-zero?

(b) What do you understand by Array Processor and Vector Processor? What is the concept of Vector Chaining?

OR

Q5' (a)' Design 16-bit adder using 4-bit CLA and compare it with16 bit adder using

→ 1-2+3-4+5...

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CPA. Also calculate the worst case add time for both? What do you understand by Carry Save Adder?

(b)' (i) What are the difference between Amdahl's law and Gustafson's Barsis law? Calculate the expected Speed up for both the law if a program consists of 0.47 fractional parts of parallel code. Which have greater speed up (if number of processor is 30) and Why? Explain with graph.

(ii) What is Super Pipelined Processor? Draw the pipeline diagram for its execution?

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Total Pages : 3

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BCEN-505

B. Tech. (Computer Engineering) Vth Semester Examination, 2014

Microprocessor

Paper No. : CEN-505

Time : 3 Hours

Maximum Marks : 60

(Write your Roll No. on the top immediately on receipt of this question paper.)

Note : ***Attempt all questions.***

1. (a) Draw and explain timing diagram of STA 2150H instruction of 8085 stored at address 2240H. **06**

   (b) Explain working of 8085 after drawing its block diagram. **06**

2. (a) (i) Explain the range of bytes that will be displayed at port 2 in the following program : **03**

```
MVI   A, BYTE 1
MOV   B, A
SUI   50H
JC    DELETE
MOV   A, B
SUI   80H
JC    Display
```


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